Day 011 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Solve Decimal Measurement Problems

Standards: 6.NS.B.3

Learning target: I can solve Decimal Measurement Problems, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9

Day 011 | Unit 01 | Lesson

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enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 43 × 8.8. Show your regrouping or division steps.

1. Estimate the size first. Treat the factors as whole numbers: 43 becomes 43, and 8.8 becomes 88.

2. Multiply the whole numbers: 43 × 88 = 3784.

3. Count decimal places: 0 in 43 plus 1 in 8.8 gives 1.

4. Place the decimal point 1 places from the right in 3784, adding leading zeros if necessary: 378.4. Remove only unnecessary trailing zeros.

Answer: 378.4

Worked example 2

Use the standard algorithm: 1.984 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 21/10
B. 31
C. 41/10
D. 31/10

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 100: 198.4 ÷ 64 = 3.1.

3. Check: 0.64 × 3.1 = 1.984.

Answer: D - 31/10

Day 011 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 77.7 × 1.9. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 9.8 ÷ 0.2. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 32.2 × 6.8. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 5 ÷ 2.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 2.37 × 7.1. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 21 ÷ 2.5. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 4.3 × 0.4. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 3.136 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 49
B. 39/10
C. 49/10
D. 59/10

09 / Multiple choice

Use the standard algorithm: 1.51 × 0.31. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 4681/1000
B. -5319/10000
C. 4681/10000
D. 14681/10000

10 / Constructed response

A container has 0.7 kg of seed. Each bag holds 0.25 kg. How many bags can be filled?

Day 011 | Unit 01 | Answer key

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1. 147.63

Compute the whole-number product and restore decimal places: 77.7 × 1.9 = 147.63.

6.NS.B.3 | Constructed response

2. 49

Multiply both numbers by 10: 98 ÷ 2 = 49. Check: 0.2 × 49 = 9.8.

6.NS.B.3 | Constructed response

3. 218.96

Compute the whole-number product and restore decimal places: 32.2 × 6.8 = 218.96.

6.NS.B.3 | Constructed response

4. 2

Multiply both numbers by 10: 50 ÷ 25 = 2. Check: 2.5 × 2 = 5.

6.NS.B.3 | Constructed response

5. 16.827

Compute the whole-number product and restore decimal places: 2.37 × 7.1 = 16.827.

6.NS.B.3 | Constructed response

6. 8.4

Multiply both numbers by 10: 210 ÷ 25 = 8.4. Check: 2.5 × 8.4 = 21.

6.NS.B.3 | Constructed response

7. 1.72

Compute the whole-number product and restore decimal places: 4.3 × 0.4 = 1.72.

6.NS.B.3 | Constructed response

8. C - 49/10

Multiply both numbers by 100: 313.6 ÷ 64 = 4.9. Check: 0.64 × 4.9 = 3.136.

6.NS.B.3 | Multiple choice

9. C - 4681/10000

Compute the whole-number product and restore decimal places: 1.51 × 0.31 = 0.4681.

6.NS.B.3 | Multiple choice

10. 2.8 bags

Multiply both numbers by 100: 70 ÷ 25 = 2.8. Check: 0.25 × 2.8 = 0.7.

6.NS.B.3 | Constructed response